

# Is There a Puzzle in Underwater Mortgage Default?

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Cleveland Fed   Boston Fed   Nebraska   Rice

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The views expressed are the authors' and do not necessarily reflect the official positions of the Federal Reserve Bank of Boston, the Federal Reserve Bank of Cleveland, or the Federal Reserve System.

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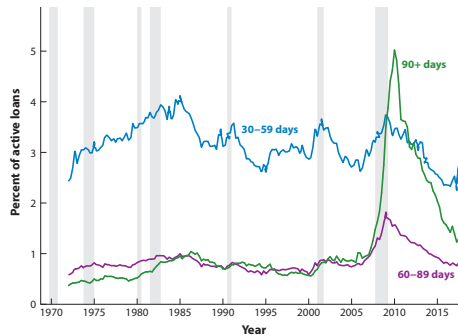
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  - ▶ Economists can’t predict the future. They can’t even predict the past.



## Motivation: the underwater default puzzle

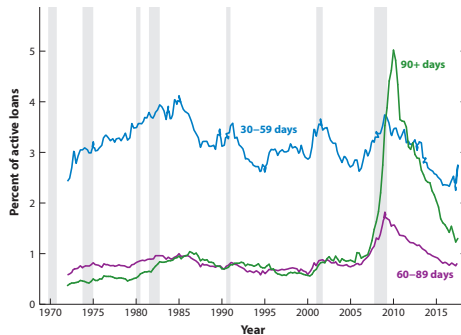
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Mortgage delinquency rates, US (Foote & Willen 2018, Fig. 1)

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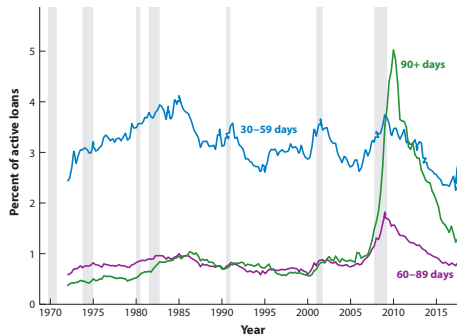
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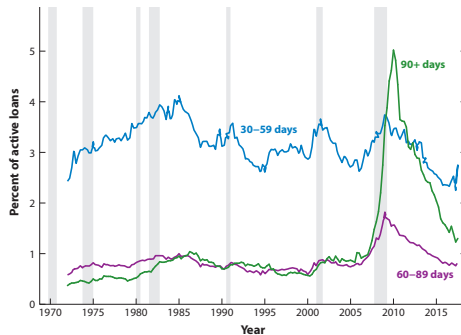
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- **This paper:** a realistic treatment of rents (and a richer choice set) **shrinks the required penalty fortyfold** — to  $0.63\%$  of lifetime consumption.



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## Where we are going

### 1. Why rent should matter

Default compares two flows, not an option on prices — and rents did not fall with prices.

### 2. The model, the calibration, and the data

Default is two decisions. A penalty worth 0.63% of lifetime consumption fits both — and an IV agrees on the magnitude.

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## The landlord's choice each period

- **Start with a landlord:** owns a property generating rent  $rent$ , financed with a mortgage of balance  $M$  and monthly payment  $r_m M$  (interest-only, for clarity).
  - ▶ **An owner-occupier faces the same arithmetic.**  
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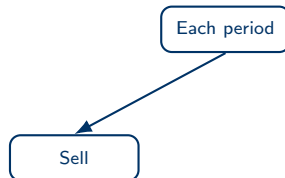
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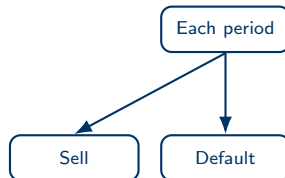
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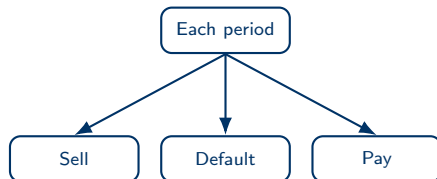
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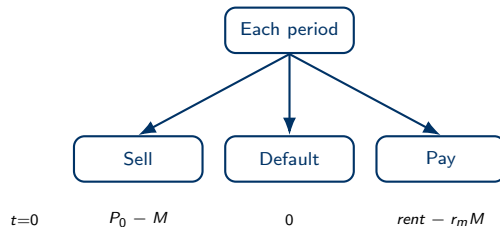
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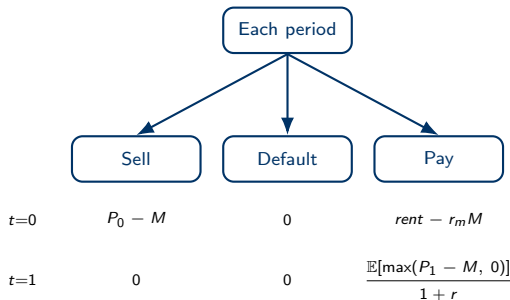
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  - ▶  $t=1$ : continuation value from holding the property and **reconsidering default next period**.

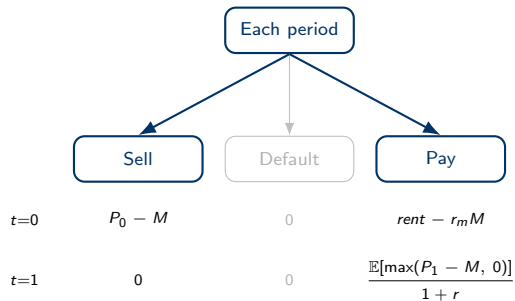


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### Positive equity ( $P_t > M$ ):

- **Selling** dominates default:  $P_0 - M > 0$ .

⇒ Default never chosen when above water.

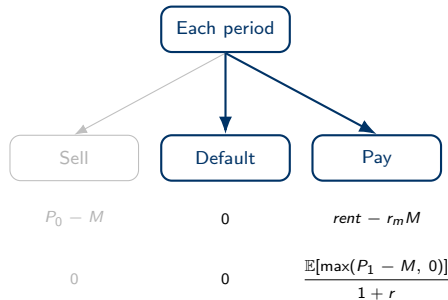


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### Negative equity ( $P_t < M$ ):

- **Default** dominates selling:  $P_0 - M < 0$ .

⇒ Real choice: default vs. pay.



## The landlord's choice each period

The default decision reduces to:

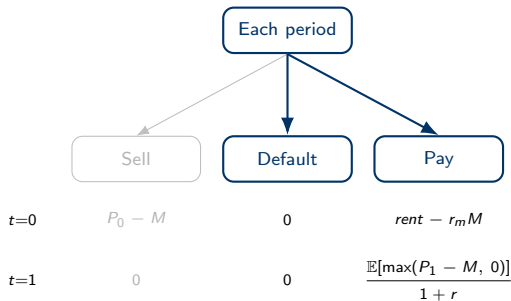
Pay if

$$\underbrace{rent - r_m M}_{\text{cash flow}_0} + \frac{1}{1+r} \underbrace{\mathbb{E}[\max(P_1 - M, 0)]}_{\text{cash flow}_1} > 0$$

or

$$\underbrace{\frac{\mathbb{E}[\max(P_1 - M, 0)]}{1+r}}_{\text{cost of default}} < \underbrace{r_m M - rent}_{\text{benefit of default}}$$

Otherwise default.

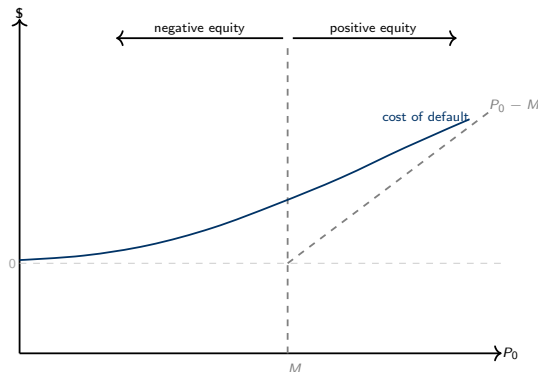


## When does the landlord default?

Default iff

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- **Cost of default** = giving up the option (blue curve). Always positive.

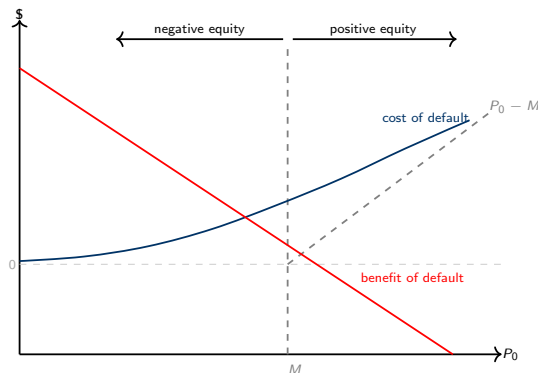


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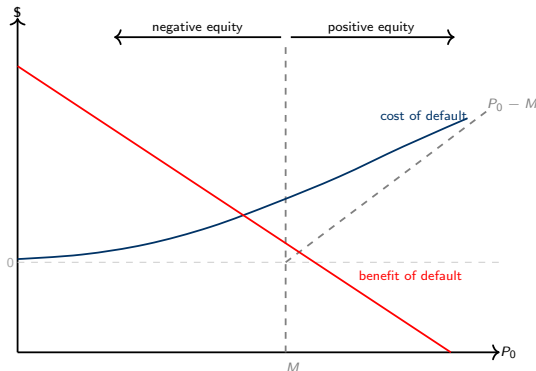


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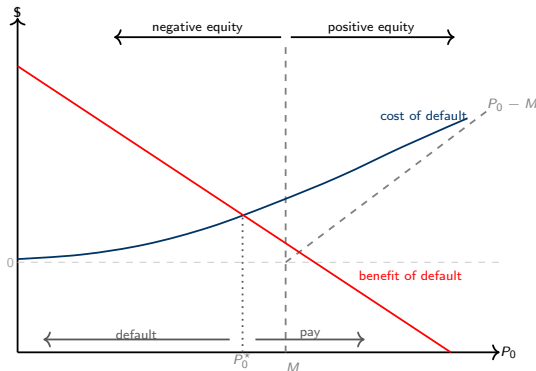


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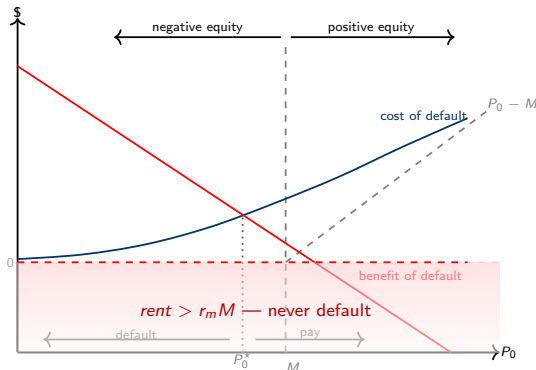


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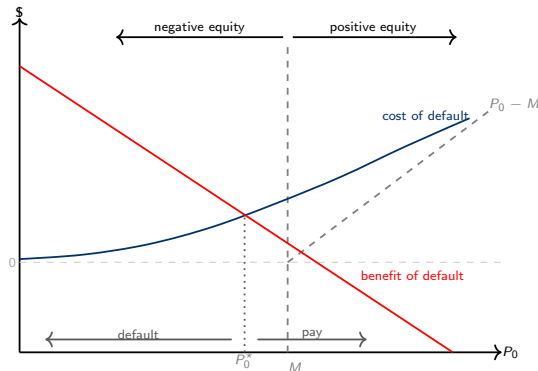
- **Special case:** if  $\text{rent} > r_m M$ , you do *not* default regardless of LTV. Keep this in mind.
- Intuitively, if you have positive cash flow you never default.

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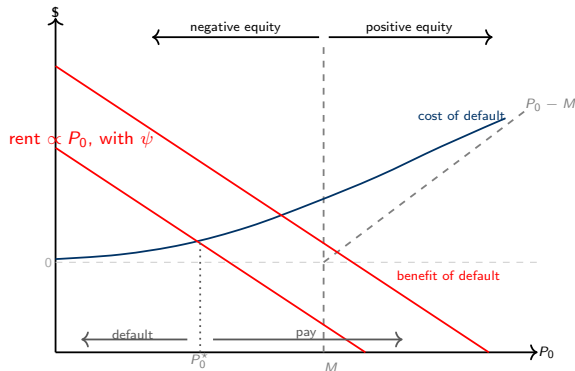


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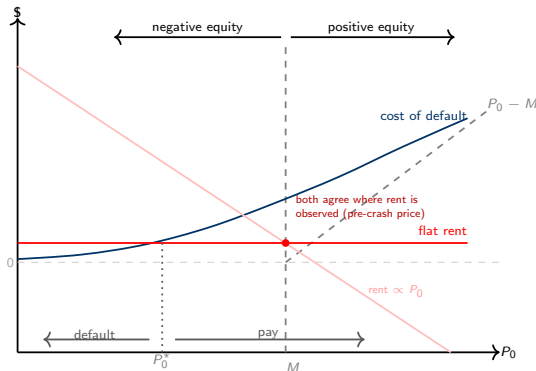


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- **Our argument:** the line should not be that steep. Both rent assumptions must match the rent we observe, at the pre-crash price — so flattening the line pivots there, and  $P_0^*$  moves far left with no penalty.



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  - ▶ price falls
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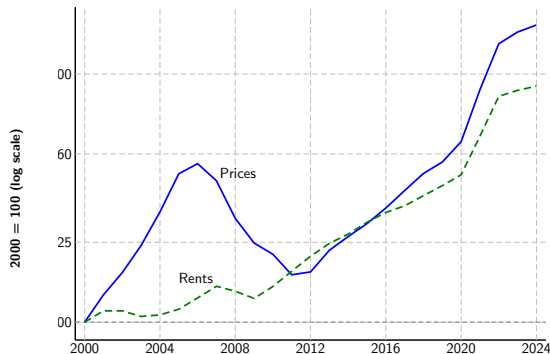
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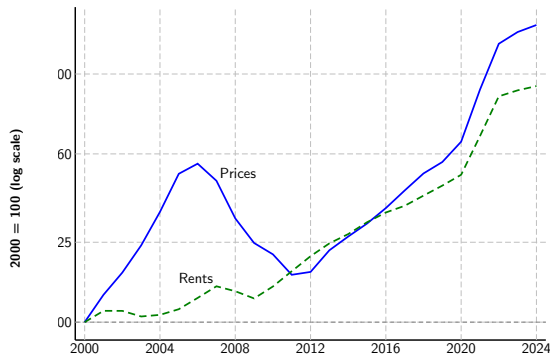
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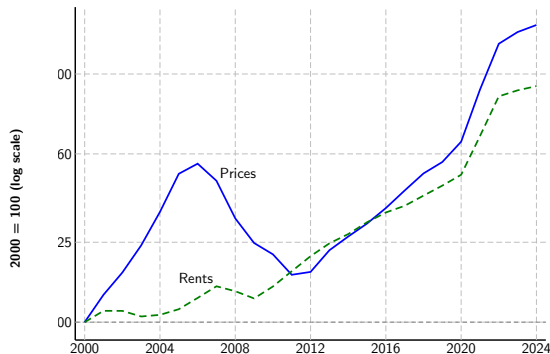
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- That is the whole intuition.
- Everything after this slide is establishing that it holds quantitatively.



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Default compares two flows, not an option on prices — and rents did not fall with prices.

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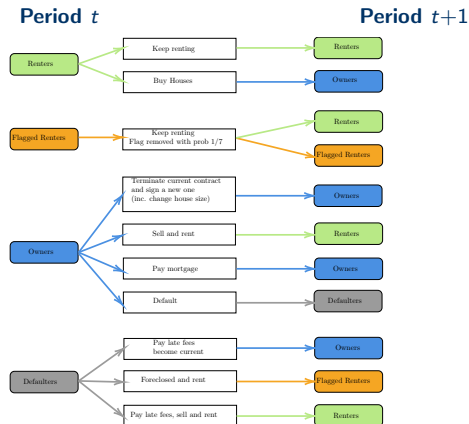
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- Endogenous tenure choice.
- The decision tree is far more complex but intuition is the same.



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**Default got every advantage we could give it. Still not worth much.**

## Calibration

- **8 parameters, 8 targets**, searched jointly

### Calibrated parameters

|            |                  | Value             |
|------------|------------------|-------------------|
| $\beta$    | Discount factor  | 0.92 <sup>†</sup> |
| $\xi$      | Substitutability | 1.98              |
| $\eta$     | Housing share    | 0.21              |
| $B_0$      | Bequest level    | 102               |
| $B_1$      | Bequest shifter  | 2.0               |
| $h_{\min}$ | Min. owned house | 0.8 <sup>†</sup>  |
| $\psi$     | Default penalty  | <b>0.13</b>       |
| $q_T$      | Distress arrival | <b>0.0011</b>     |

<sup>†</sup> At a search-box edge; both bind toward *more* default, so the fit is not flattered.

# Calibration

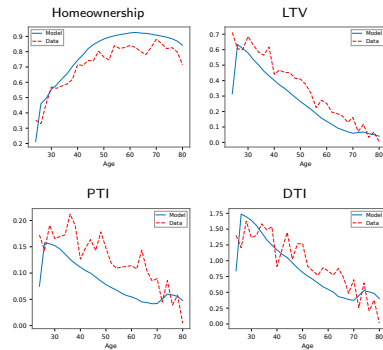
- 8 parameters, 8 targets, searched jointly
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## Calibrated parameters

|            |                  | Value            |
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| $\beta$    | Discount factor  | $0.92^{\dagger}$ |
| $\xi$      | Substitutability | 1.98             |
| $\eta$     | Housing share    | 0.21             |
| $B_0$      | Bequest level    | 102              |
| $B_1$      | Bequest shifter  | 2.0              |
| $h_{\min}$ | Min. owned house | $0.8^{\dagger}$  |
| $\psi$     | Default penalty  | <b>0.13</b>      |
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## Targets I: life-cycle profiles



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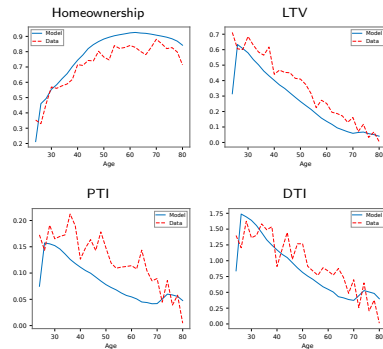
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### Targets II: delinquency moments

|                                       | Model | Data  | Fit   |
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| Delinquency by LTV, 15 bins           | —     | —     | 0.067 |
| Above-water delinquency, %/yr         | 2.11  | 2.03  | 0.002 |
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| Above-water foreclosure   delinquency | 0.375 | 0.402 | 0.005 |

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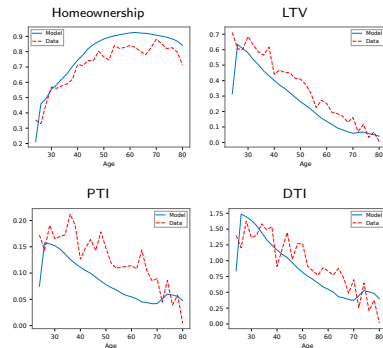
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- **Gauss–Newton** on normalized squared discrepancies
  - ▶ Converged iteration 9, objective **0.239**
- $\psi = 0.13$  in utility units
  - ▶ **0.63% of lifetime consumption** (CEV)
- **Model LTV measured as the data measure it**
  - ▶ Entry-year balance, log-error s.d. 0.14
- **No ownership premium**
  - ▶ Discipline is unit-level rent-to-price (Loewenstein 2023)

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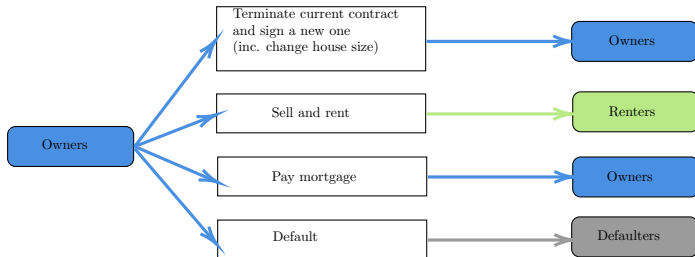
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- **Stage 1 — an owner stops paying**

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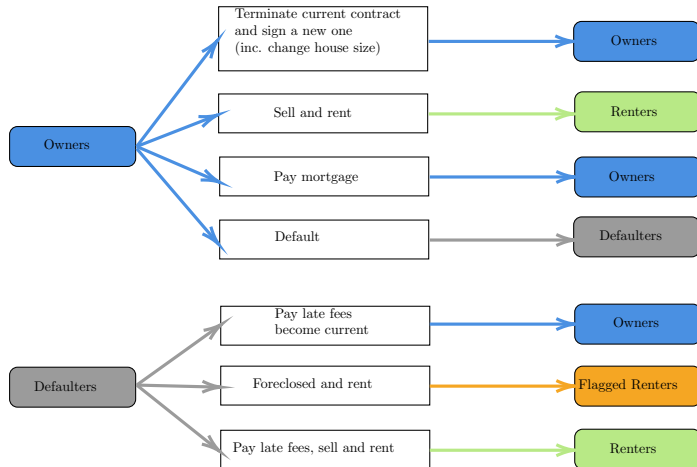
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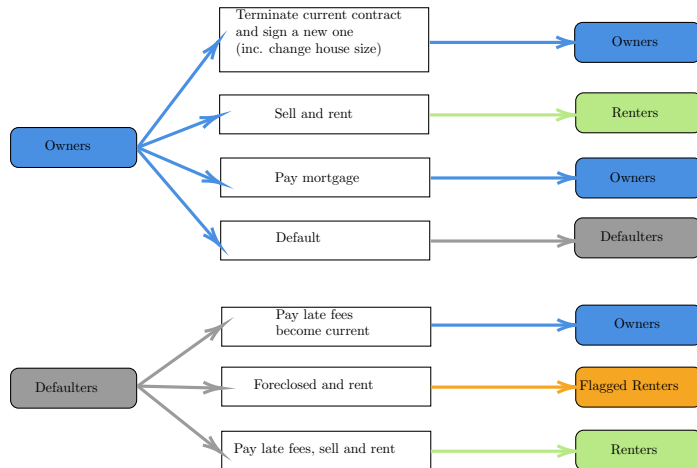
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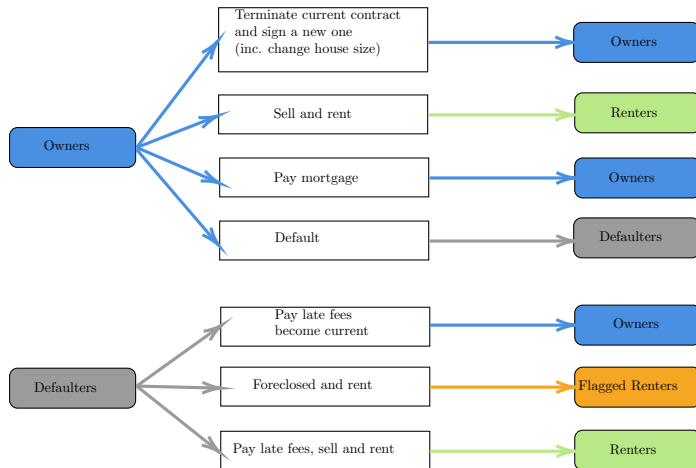
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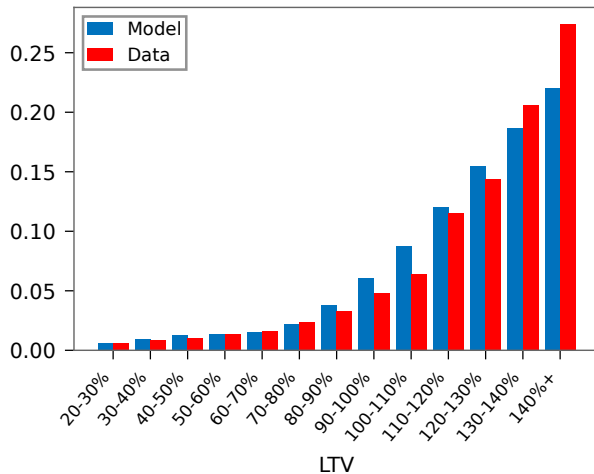
- **So we show the two stages separately**

- ▶ Who enters (next), then how it ends
- ▶ Above-water default lives in *both*



## Default by LTV: model versus data

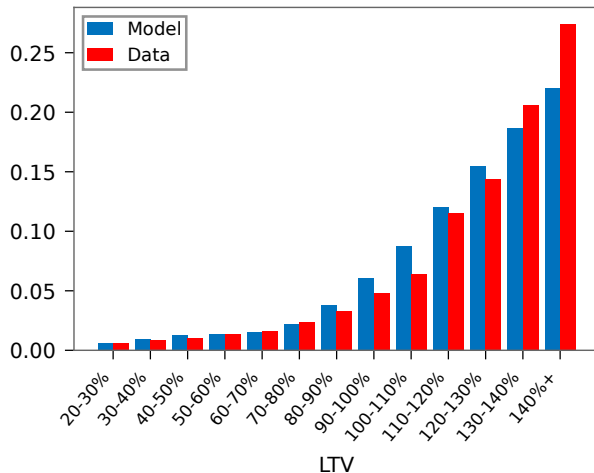
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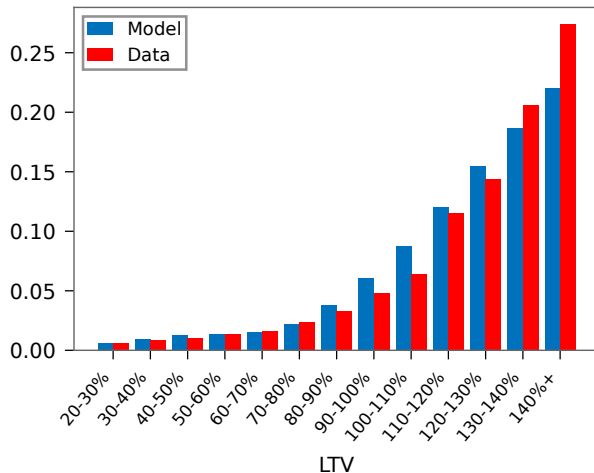
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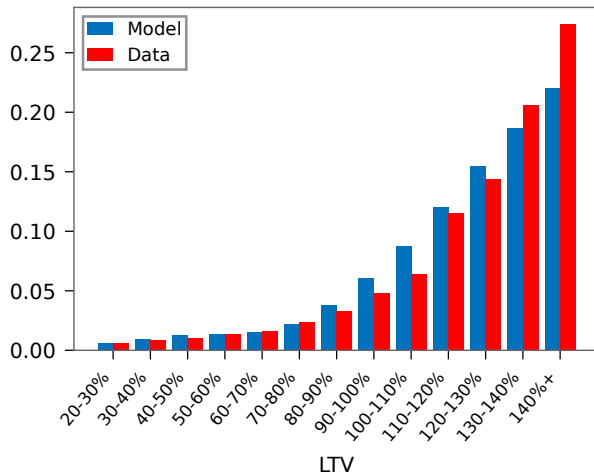
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- **Above-water bars are positive — and should be**
  - ▶ Stage 1 is missing a payment, not losing the house
  - ▶ A liquidity event, available at any LTV
  - ▶ Nothing here is strategic

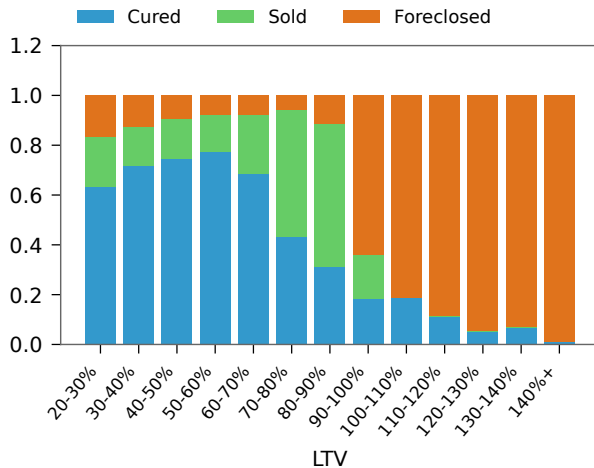


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## After default: cure, sell, or foreclose

### • Stage 2: how delinquency ends

- ▶ Above water, **curing dominates** (peak 0.78)
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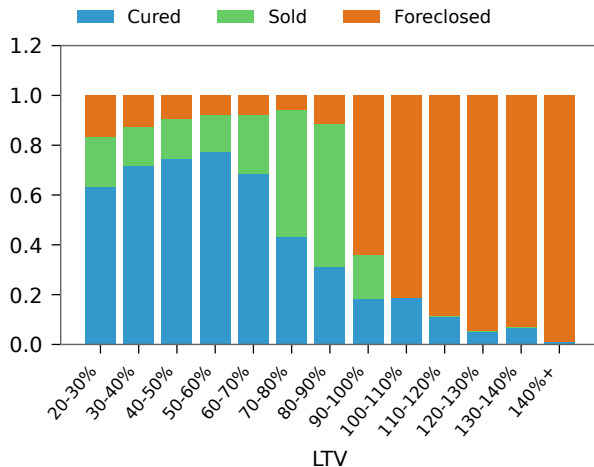
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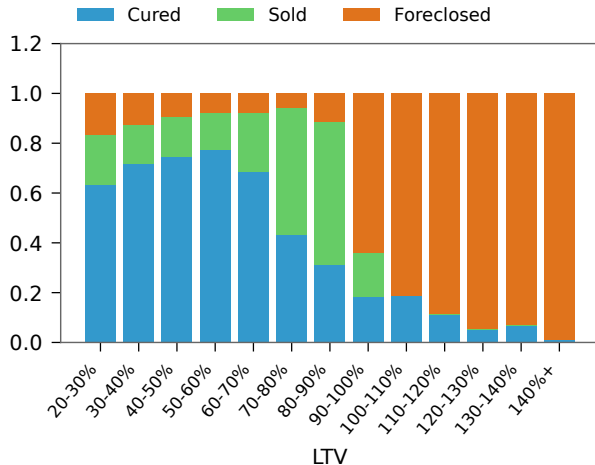
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- **None of the three is a taste for default**



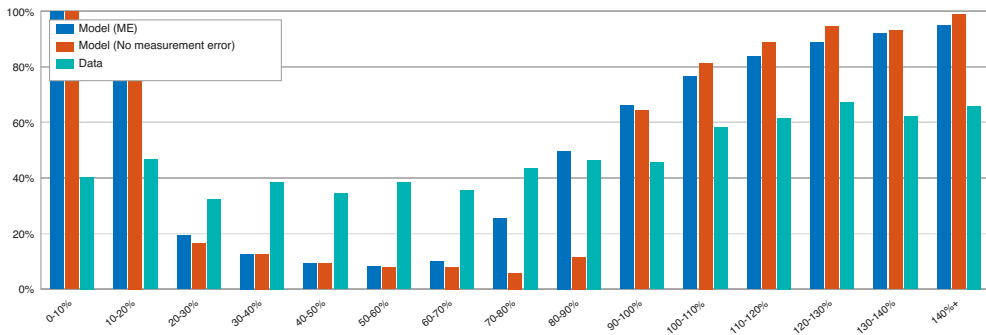
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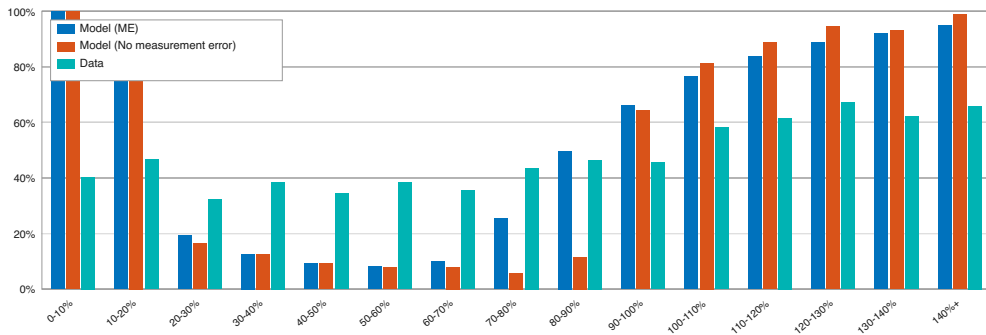
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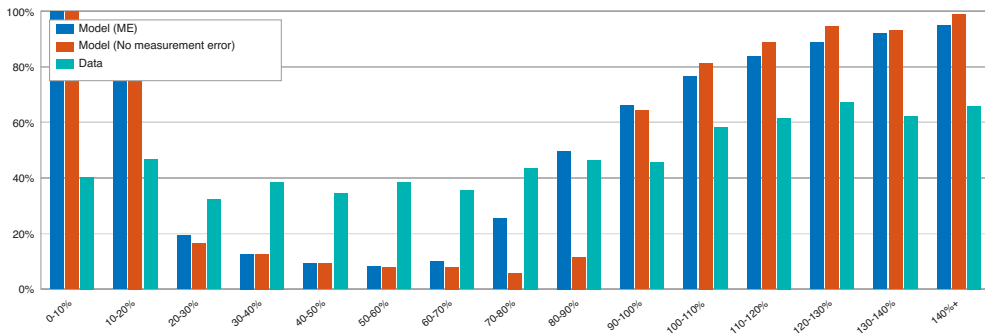
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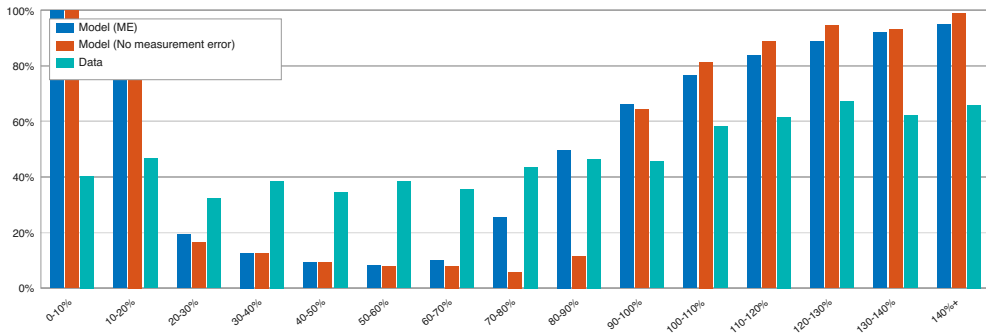
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- ▶  $\psi$  is fit to the *aggregate* above-water rate
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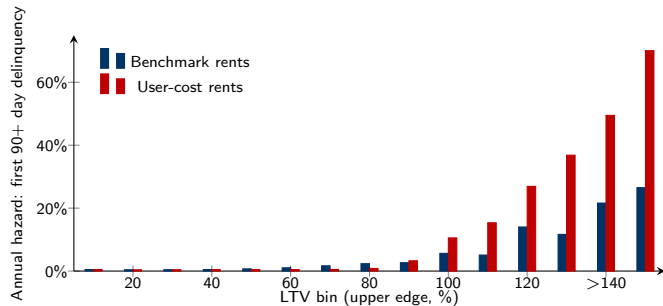
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## User-cost rents, our model

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Transition off the calibrated stationary equilibrium: rent falls unexpectedly, ownership and leverage inherited — a Campbell–Cocco-style response, not a re-solved steady state. First year shown; years 2–5 within a few points. Aggregate delinquency 2.4%  $\rightarrow$  3.3–3.9%; homeownership 75%  $\rightarrow$  44%.

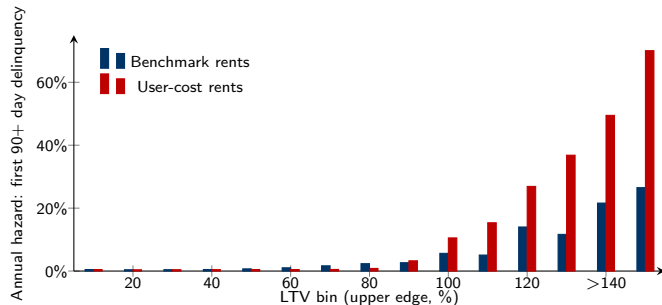
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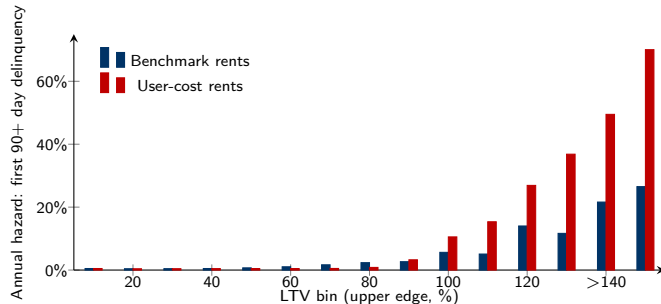
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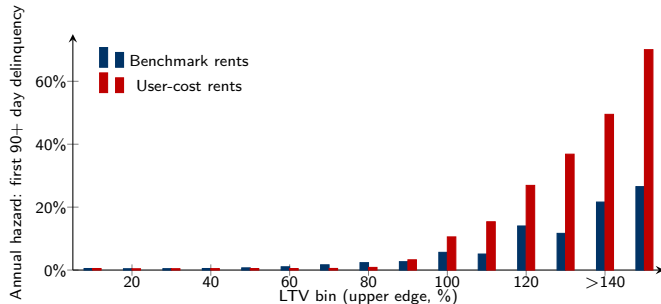
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|                           | (1)                     | (2)                     | (3)                     |
|---------------------------|-------------------------|-------------------------|-------------------------|
|                           | OLS                     | OLS                     | IV                      |
| $\Delta \ln(\text{Rent})$ | $-0.86^{***}$<br>(0.06) | $-0.65^{***}$<br>(0.06) | $-2.86^{***}$<br>(0.72) |
| Current LTV               | $5.44^{***}$<br>(0.04)  | $5.40^{***}$<br>(0.04)  | $5.19^{***}$<br>(0.08)  |
| $\ln(\text{Payment})$     | $-3.48^{***}$<br>(0.03) | $-3.38^{***}$<br>(0.03) | $-3.39^{***}$<br>(0.03) |
| Fixed rate                | $-3.36^{***}$<br>(0.06) | $-3.49^{***}$<br>(0.06) | $-3.46^{***}$<br>(0.06) |
| $\ln(\text{Avg. wages})$  |                         | $-0.79^{***}$<br>(0.03) | $-0.70^{***}$<br>(0.05) |
| Unemployment              |                         | $-0.12^{**}$<br>(0.06)  | $-0.58^{***}$<br>(0.16) |
| Adj. $R^2$                | 0.052                   | 0.052                   | 0.014                   |
| Local controls            |                         | Y                       | Y                       |
| Year, orig., state FE     | Y                       | Y                       | Y                       |
| First-stage $F$           |                         |                         | 18,200                  |

$N = 2,511,128$  in all columns.

**Notes.** Dependent variable  $100 \times \mathbb{I}\{\text{foreclosure completed}\}$ , mean 25.4. One observation per loan at its first 90-day default. All regressors standardized;  $\Delta \ln(\text{rent})$  has s.d. 8.4 log points, so column (2) is  $-0.82$  pp and column (3)  $-3.59$  pp per 10% rent decline. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ . Paper table *Probability of Foreclosure Conditional on 90-day Default*, cols. (1), (3), (4).

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- **Why condition on default?** Everyone here has a liquidity problem. What varies is the cost of walking away.
- **IV (Gete–Reher 2018):** CBSA stress-tested-bank share.  $\text{IV} > \text{OLS}$ , so OLS is attenuated.

$$D_i = \beta_1 \Delta \ln(\text{rent}_{o,t(i)}) + \beta_2 \text{LTV}_i + \zeta X_i + \gamma_{t(i)} + \lambda_{o(i)} + \delta_i + \epsilon_i$$

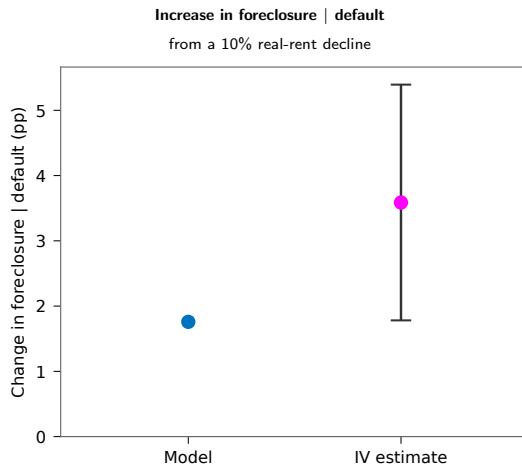
|                           | (1)                | (2)                | (3)                |
|---------------------------|--------------------|--------------------|--------------------|
|                           | OLS                | OLS                | IV                 |
| $\Delta \ln(\text{Rent})$ | −0.86***<br>(0.06) | −0.65***<br>(0.06) | −2.86***<br>(0.72) |
| Current LTV               | 5.44***<br>(0.04)  | 5.40***<br>(0.04)  | 5.19***<br>(0.08)  |
| $\ln(\text{Payment})$     | −3.48***<br>(0.03) | −3.38***<br>(0.03) | −3.39***<br>(0.03) |
| Fixed rate                | −3.36***<br>(0.06) | −3.49***<br>(0.06) | −3.46***<br>(0.06) |
| $\ln(\text{Avg. wages})$  |                    | −0.79***<br>(0.03) | −0.70***<br>(0.05) |
| Unemployment              |                    | −0.12**<br>(0.06)  | −0.58***<br>(0.16) |
| Adj. $R^2$                | 0.052              | 0.052              | 0.014              |
| Local controls            |                    | Y                  | Y                  |
| Year, orig., state FE     | Y                  | Y                  | Y                  |
| First-stage $F$           |                    |                    | 18,200             |

$N = 2,511,128$  in all columns.

**Notes.** Dependent variable  $100 \times \mathbb{I}\{\text{foreclosure completed}\}$ , mean 25.4. One observation per loan at its first 90-day default. All regressors standardized;  $\Delta \ln(\text{rent})$  has s.d. 8.4 log points, so column (2) is −0.82 pp and column (3) −3.59 pp per 10% rent decline. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ . Paper table *Probability of Foreclosure Conditional on 90-day Default*, cols. (1), (3), (4).

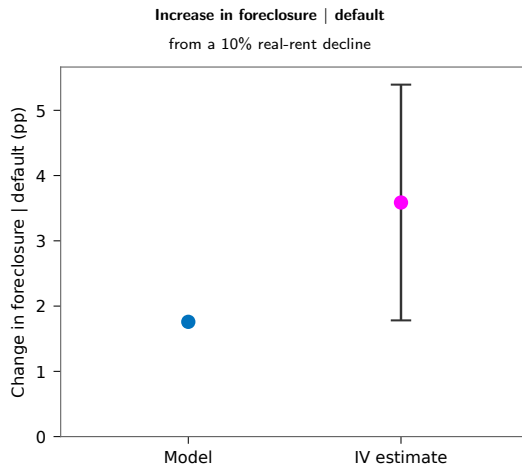
## Model versus IV

- **One number, measured two ways**
  - ▶ Extra foreclosure per delinquent borrower
  - ▶ From a 10% cheaper outside option



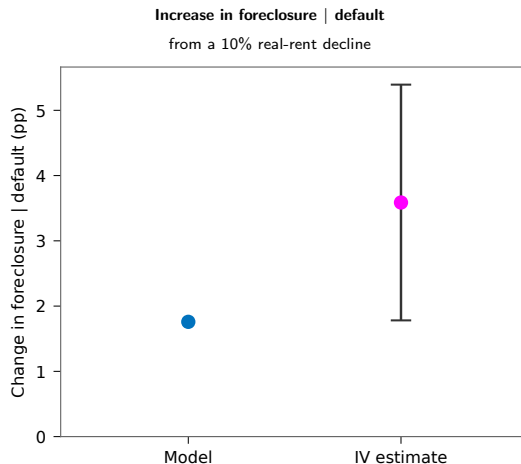
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- **Left bar: the model**
  - ▶ Cut real rent 10%: 46.22 → 47.98%
  - ▶ **1.76 pp**



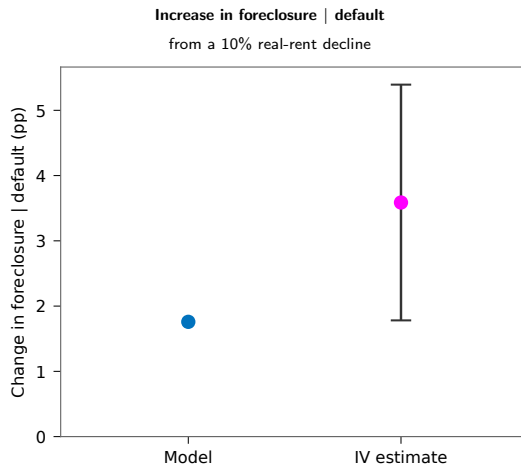
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- **Right bar: the IV**
  - ▶ Same 10% decline: **3.59 pp**, s.e. 0.90
  - ▶ Whiskers  $\pm 2$  s.e.: [1.78, 5.39]



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  - ▶ Model at the low end of the band
  - ▶ The structural number is the conservative one



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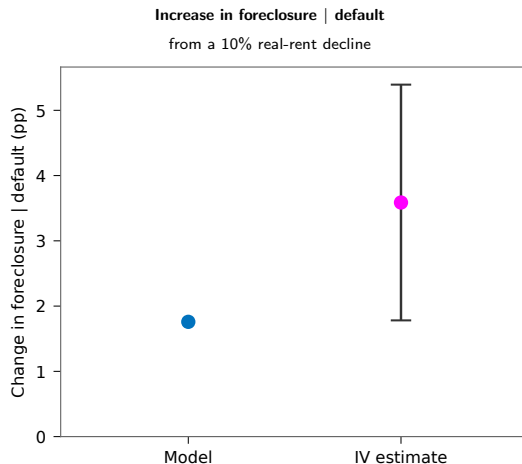
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- **Same sign, same order of magnitude**

- ▶ Model at the low end of the band
- ▶ The structural number is the conservative one

- **And never a calibration target**

- ▶ A life-cycle model and a bank instrument agree
- ▶ A model that fits vs. the right mechanism



## 1. Why rent should matter

Default compares two flows, not an option on prices — and rents did not fall with prices.

## 2. The model, the calibration, and the data

Default is two decisions. A penalty worth 0.63% of lifetime consumption fits both — and an IV agrees on the magnitude.

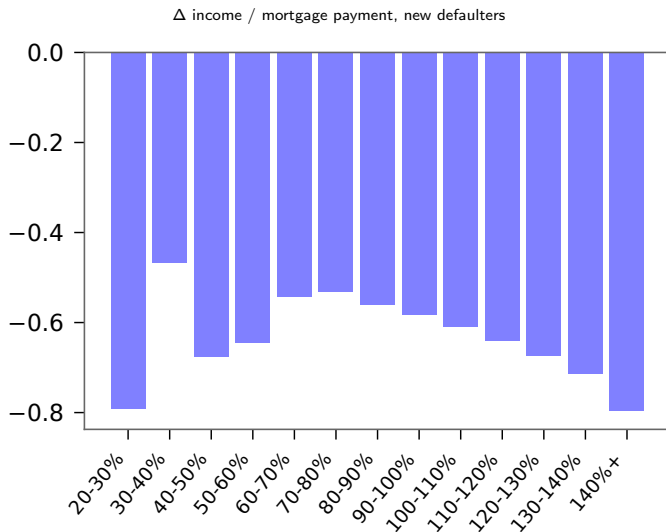
## 3. Questions and answers

Campbell–Cocco, the Ganong–Noel fact, where we fail, policy.

## Ganong–Noel (2023)

- **Their diagnostic: the income drop at default**

- ▶ Scaled by the monthly mortgage payment
- ▶ They find declines near one payment



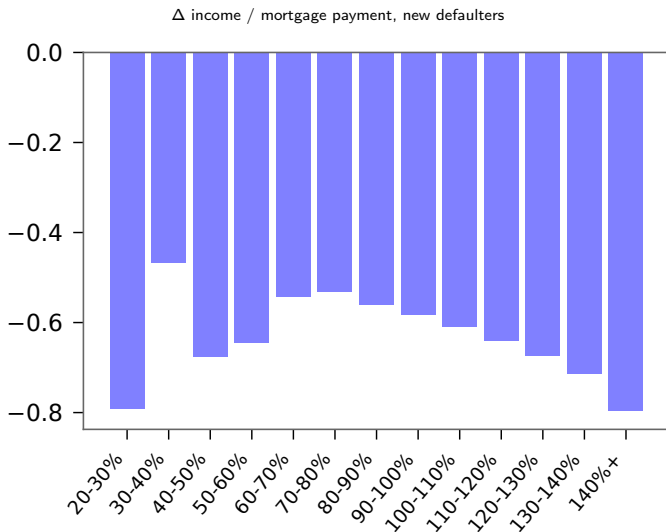
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- **Our model generates it**

- ▶ Income falls before default at every LTV
- ▶  $-0.47$  to  $-0.80$  of a payment
- ▶  $-0.64$  above water,  $-0.82$  below



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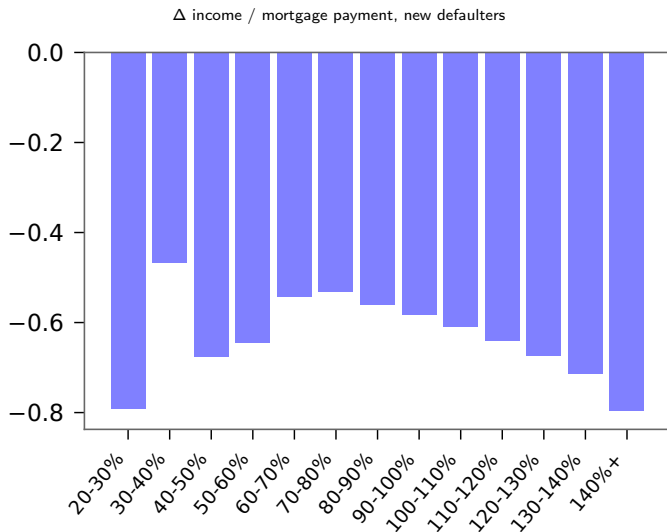
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- **Not a target**

- ▶ Nothing here was fit to an income profile



## Campbell–Cocco (2015)

- **Not our model — theirs**

- ▶ Where the large penalty was estimated
- ▶ Fit to the income declines we just saw
- ▶ Vary two things: rent process, penalty

**Campbell–Cocco model: does it match  
G&N's income declines at default?**

|                     |                         |
|---------------------|-------------------------|
| No penalty          | $\psi \approx 25\%$ CEV |
| Rents fall with $p$ |                         |
| Constant real rent  |                         |

Paper Appendix Figure A.1, Panels (a)–(d).

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- **The rent assumption was the penalty**

- ▶ Not a distaste for default
- ▶ An unrealistic rent process, absorbed

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A large  $\psi$  also makes the model *insensitive* to flows. Cut rent 10% permanently in their high-penalty baseline and default does not budge: **0.19% → 0.19%** (Appendix Table A.1).

## Policy: forbearance

- **One condition: get the payment under the rent**
  - ▶ The short-run benefit of default is  $r_m M - \text{rent}$
  - ▶ Postponement sets the payment to zero
  - ▶ So the benefit is *negative* — nobody walks

### Default rate by equity position

paper table *Default and Foreclosure Outcomes: Abovewater vs. Underwater*, Panels A–C

|                         | Above water | Underwater  |
|-------------------------|-------------|-------------|
| Data (CRISM)            | 2.03        | 10.12       |
| Baseline model          | 2.11        | 9.85        |
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### Aggregate effect

paper table *Default and Foreclosure Outcomes: Mortgage Forbearance*

|                        | Default     | Foreclosure |
|------------------------|-------------|-------------|
| Baseline               | 2.35        | 1.09        |
| Stationary eq. with FB | 2.16        | 1.06        |
| Year of introduction   | 1.28        | 0.98        |
| <b>One year after</b>  | <b>1.48</b> | <b>0.46</b> |

Forbearance: one-year payment postponement for mortgagors with PTI > 31%.  
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- **Insurance against a shock, not a lower default rate**

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## Takeaways

- **Rent should matter, intuitively.** Default is a comparison of flow costs, not an option on house prices. Rents did not fall with prices, so the “puzzle” is an artifact of assuming they did.
- **Rent matters quantitatively, in the model.** With realistic rents and a realistic choice set — downsizing, post-default rebuying, a free year of housing — the penalty needed is **0.63%** of lifetime consumption, against 25% (Campbell–Cocco) and 29% (Laufer).
- **And by about the same amount in the data.** The flow-cost elasticity is not a calibration target, yet the model’s **1.76 pp** sits at the low edge of the two-s.e. band of an IV built from bank stress tests (**3.59 pp**, band [1.78, 5.39]).
- *Policy:* cash-flow relief is the effective lever, but forbearance is insurance against a shock, not a permanent cut in default — anticipating households lever up.
- *A caution for empirical work:* the Ganong–Noel strategic share is not identified without an assumption about the above-water comparison group. Inside our model we move it from 39% to zero while underwater default behavior is completely unchanged.

## So why keep a penalty at all?

- Realistic rents shrink the penalty; they do not eliminate it. Ours is  $\psi = 0.13$ , **worth 0.63% of lifetime consumption** — fortyfold below the literature, but not zero.

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| Limited rental after fcl. | 1.71        | 4.92         |
| ... and <i>no</i> penalty | <b>9.55</b> | <b>19.22</b> |

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- **But take the penalty away entirely and the model breaks** — default overshoots the data badly in *both* groups, 9.55% above water against 2.03% in CRISM.
- **Reading:** the puzzle is not that the penalty is implausibly large. It is that the earlier models had to make it large.

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## On request: does this travel to Canada?

- Our mechanism is about **rents, the choice set, and liquidity** — none of which is US-specific. But the institutions differ, and they push in identifiable directions.

| Feature       | US                 | Canada          |
|---------------|--------------------|-----------------|
| Rate term     | 30-yr fixed        | 5-yr, renewed   |
| Recourse      | Varies by state    | Mostly recourse |
| High-LTV ins. | FHA / PMI          | CMHC            |
| Prepayment    | Free               | Penalized       |
| Rent dynamics | Loewenstein (2023) | ?               |

**The question for this room — the last row.** We do not know whether Canadian rents track Canadian prices the way the models assume. If they do, the puzzle here is real and our resolution does not travel. If they do not, it should. Which of these institutions makes measured above-water default larger or smaller than ours?

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- **Recourse** raises the effective cost of default without any nonpecuniary penalty. Less need for  $\psi$ , not more.
- **Five-year terms with renewal** put a scheduled liquidity shock in every borrower's path. Our liquidity channel should be *stronger*, and predictable in timing.

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- **Recourse** raises the effective cost of default without any nonpecuniary penalty. Less need for  $\psi$ , not more.
- **Five-year terms with renewal** put a scheduled liquidity shock in every borrower's path. Our liquidity channel should be *stronger*, and predictable in timing.
- **CMHC insurance at high LTV** changes who bears the loss, so the lender-NPV calculus behind forbearance is a different calculation — but the borrower's problem is the same one.

| Feature       | US                 | Canada          |
|---------------|--------------------|-----------------|
| Rate term     | 30-yr fixed        | 5-yr, renewed   |
| Recourse      | Varies by state    | Mostly recourse |
| High-LTV ins. | FHA / PMI          | CMHC            |
| Prepayment    | Free               | Penalized       |
| Rent dynamics | Loewenstein (2023) | ?               |

**The question for this room — the last row.** We do not know whether Canadian rents track Canadian prices the way the models assume. If they do, the puzzle here is real and our resolution does not travel. If they do not, it should. Which of these institutions makes measured above-water default larger or smaller than ours?

## On request: the Ganong–Noel strategic share

- Every default here is a double trigger
  - ▶ More equity would reduce it (0.4% → 26.5% across LTV)
  - ▶ More income would reduce it, at every LTV
  - ▶ **No equity-only group, no income-only group**

### G&N's income comparison

$$\hat{\theta}^{GN} = \max\{0, 1 - (\mu_U - \mu_0)/(\mu_A - \mu_0)\}, \mu_0 = 0.475$$

| Scenario               | $\Delta y$ / payment |        | G&N-implied<br>UW strategic share |
|------------------------|----------------------|--------|-----------------------------------|
|                        | AW                   | UW     |                                   |
| Baseline               | −0.636               | −0.820 | 0.0%                              |
| + UW default shock     | −0.636               | −0.201 | <b>39.1%</b>                      |
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AW/UW split on true LTV at 100%, ages 24–79. Share floored at zero. Applies G&N's formula to annual model moments, not their monthly estimator. Paper table *Abovewater Default Selection and Implication for G&N*.

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- And it is fragile
  - ▶ Start where it returns something (row 2 — baseline gives zero)
  - ▶ Shock the above-water group *only*: 39% → 0
  - ▶ Underwater behavior unchanged

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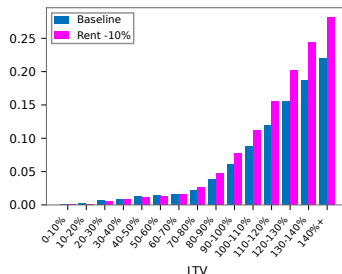
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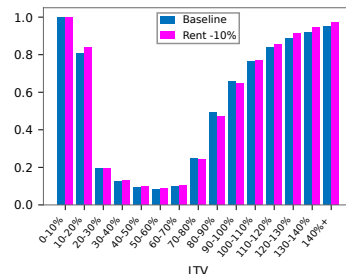
**Underwater behavior is identical in rows 2 and 3. The share moves 39% → 0 anyway — it is a statement about the comparison group.**

## Counterfactual: a permanent 10% decline in real rent

- The cleanest test of the mechanism *inside* the model: make the outside option 10% cheaper, change nothing else. That is about the largest real-rent decline borrowers actually saw in 2007–2012 — though ex post that one was not permanent.



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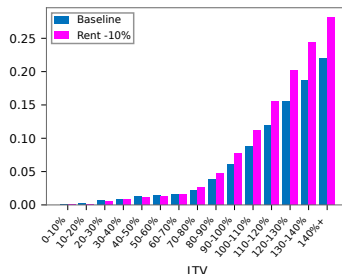


(b) Foreclosure rate | default

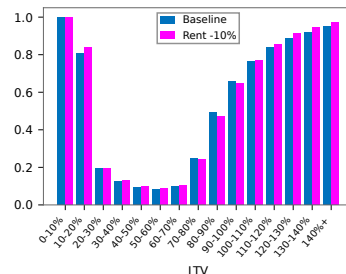
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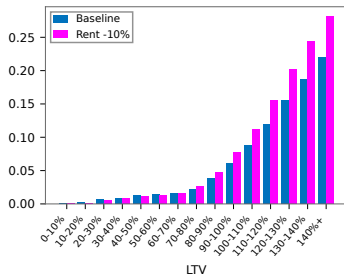


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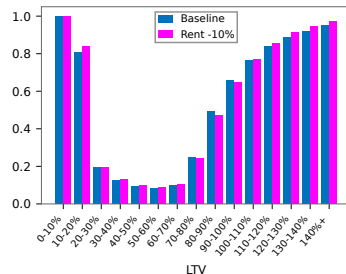
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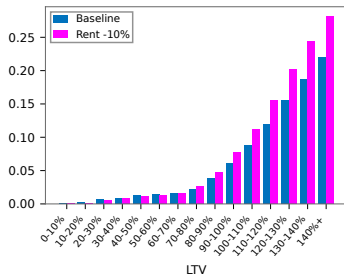


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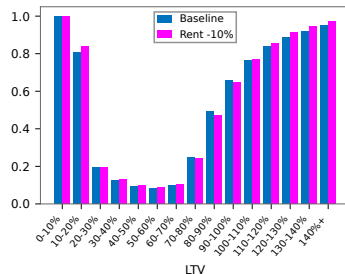
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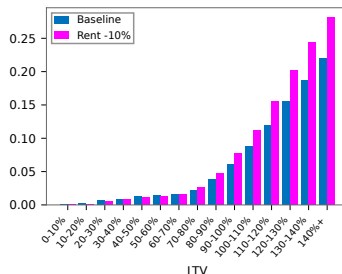


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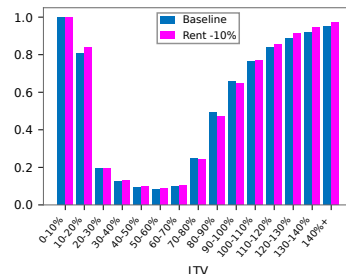
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- Negative equity is not what creates the incentive — the **flow cost of the alternative** is. Hence a **triple trigger**: equity, income, and flow costs.



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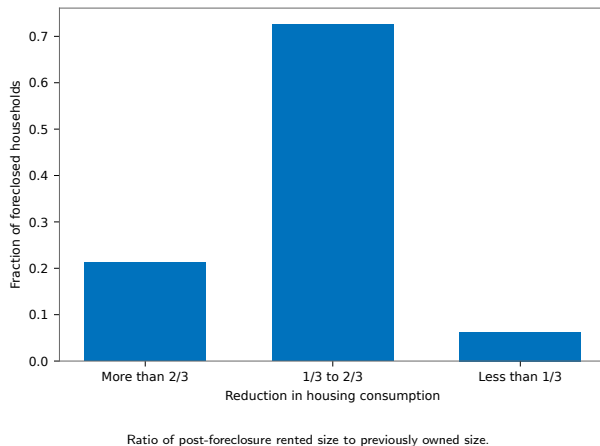


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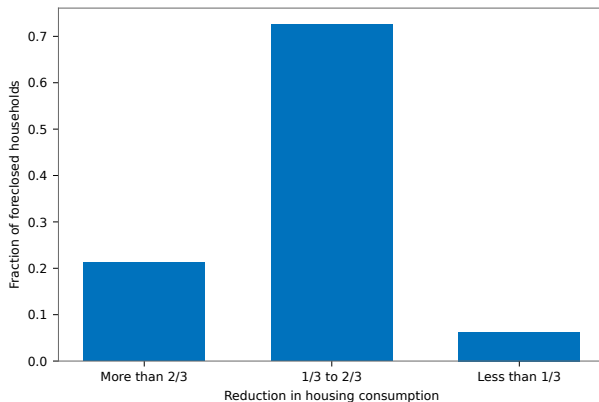
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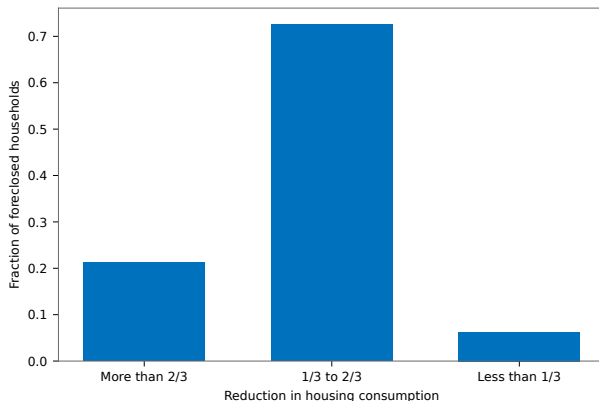
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Ratio of post-foreclosure rented size to previously owned size.

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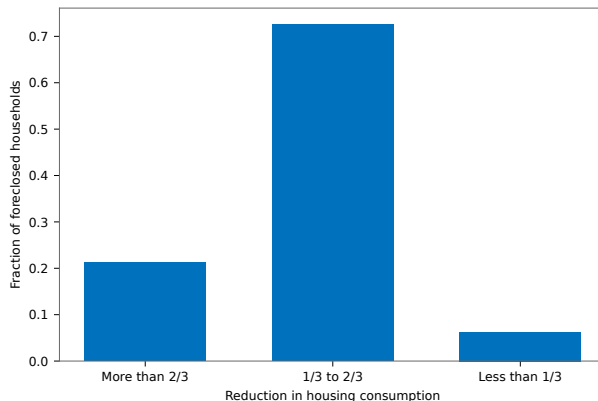
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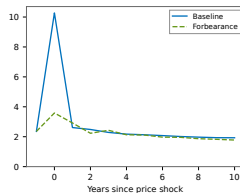
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- This is why the flow saving from walking away is real — and why we need only a small  $\psi$  to match the data.



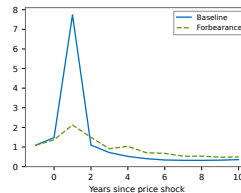
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## Forbearance under a house-price crash

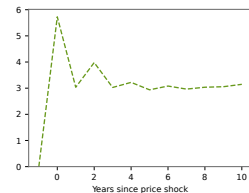
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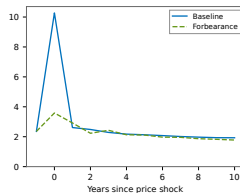
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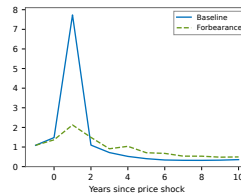
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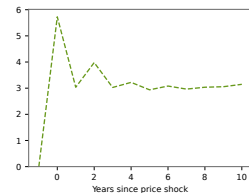
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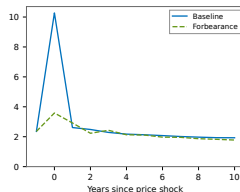
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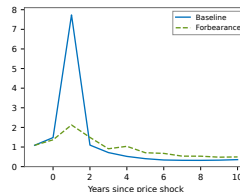
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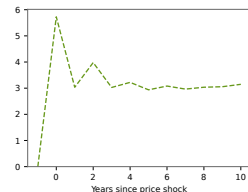
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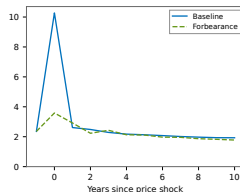
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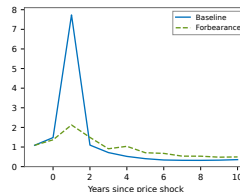
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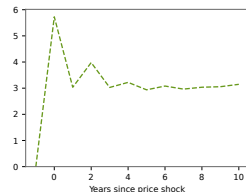
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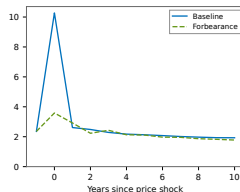
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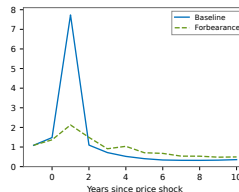
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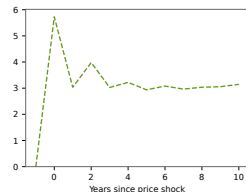
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- **Take-up peaks at 8%** and settles near 5% — a small, self-limiting program.
- Costless to a risk-neutral lender; *valuable* to one whose marginal utility is high in a crash.



(a) Delinquency



(b) Foreclosure



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- A small speculative group may be exactly why our baseline income gradient is steeper than G&N's data.
- A 30% lower rent-to-price paired with a 30% ownership premium gives *the same* household choices as our calibration.

## Three things to take away

- 1. Rent should matter — intuitively

- ▶ Default compares two flows, not an option on prices
- ▶ Rents did not fall with prices
- ▶ So walking away was never as cheap as assumed

**Intuition**

rents  $\neq$  prices, so the  
default option is worth less

**Model**

small  $\psi$  (0.63% CEV)  
fits the default data

**Data**

IV on rent growth agrees  
on the magnitude

**Warning**

G&N  $\hat{s}$  hinges on the  
above-water group: 39%  $\rightarrow$  0

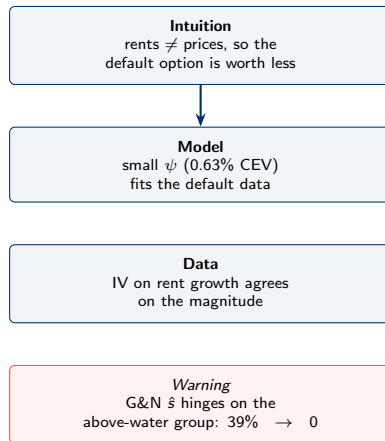
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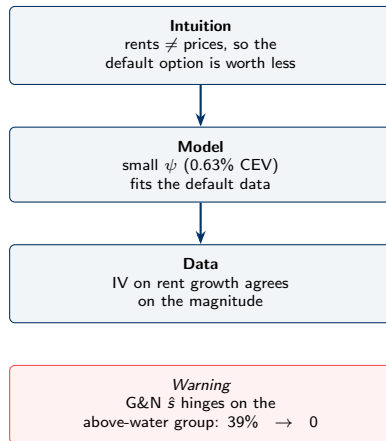
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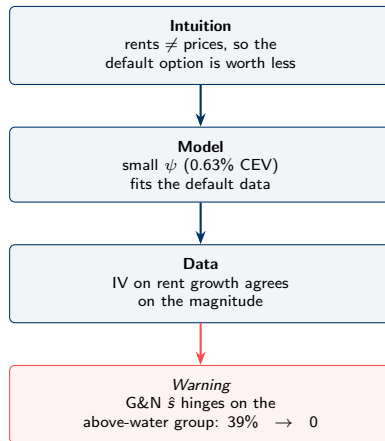
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- **3. And by about the same amount in the data**
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  - ▶ Model 1.76 pp, at the low edge of the IV band



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- **1. Rent should matter — intuitively**
  - ▶ Default compares two flows, not an option on prices
  - ▶ Rents did not fall with prices
  - ▶ So walking away was never as cheap as assumed
- **2. Rent matters quantitatively — in the model**
  - ▶ Required penalty: **0.63%** of lifetime consumption
  - ▶ Against 25% (Campbell–Cocco), 29% (Laufer)
- **3. And by about the same amount in the data**
  - ▶ Flow-cost elasticity is *not* a calibration target
  - ▶ Model 1.76 pp, at the low edge of the IV band
- *And on Ganong–Noel: we match their fact*
  - ▶ Every default here is a *double trigger*
  - ▶ So the equity-only share has no clean target
  - ▶ We move it 39%  $\rightarrow$  0, underwater behavior fixed



## Which ingredient does the work? (within our model)

- We now have the rent row in *our* model — user-cost rents put the puzzle back. What we do not have is how much the *choice set* contributes on its own.

| Switch off . . .                                | Deep-UW hazard            | vs data 27%      |
|-------------------------------------------------|---------------------------|------------------|
| nothing (baseline)                              | 26.5%                     | matches          |
| <b>rent realism</b> ( $r \downarrow$ with $p$ ) | <b>70.0%</b>              | <b>over 2.6×</b> |
| <i>And on the penalty itself:</i>               |                           |                  |
| $\psi$ entirely, options intact                 | AW default 9.55% vs 2.03% |                  |

**What we can defend:** rent realism alone brings the puzzle back, with every generous option still switched on — so the rent process is sufficient.

**What we cannot yet say:** how much the choice set contributes on its own. Turning off downsizing, the rebuy, the free year or  $k_s$  one at a time and re-fitting  $\psi$  needs four solves we do not have. The attribution is one-sided by construction, and I would rather say that than guess.

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- The exercise: switch off one ingredient, hold the other seven parameters at their calibrated values, and see what happens to the moment  $\psi$  was fit to.
- **Rent realism alone is enough to break the fit.** What we cannot yet say is whether the choice set — downsizing, post-foreclosure rebuy, the free year — would break it too.

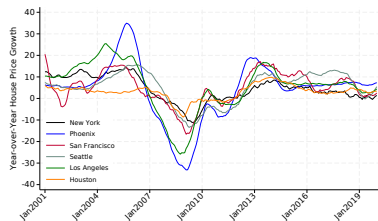
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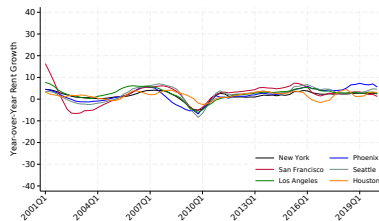
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## Backup: the same picture city by city

- The headline figure is a national index. The obvious worry is that it averages away the bust markets — Phoenix, Las Vegas, the Central Valley — where the underwater borrowers actually were.
- It does not. Rents fell in those cities too, but by **roughly an order of magnitude less than prices**: house-price growth troughs near  $-30\%$  year over year, rent growth near  $-5$  to  $-10\%$ .
- So the flow comparison moves the way the national picture says it does, in exactly the places where the puzzle is measured.



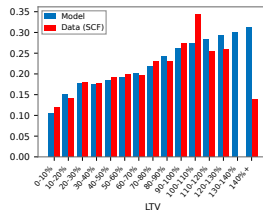
(a) House-price growth, CoreLogic HPI



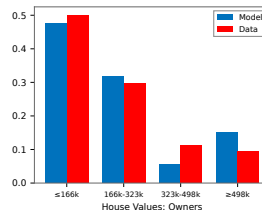
(b) Effective market rent growth, CoStar

## Backup: fit in *nontargeted* moments

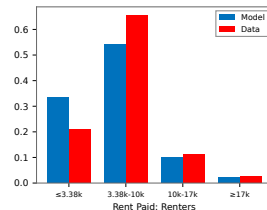
- Nothing here was targeted. The comparison is to the **2001 SCF**.
- Payment-to-income by LTV, the house-value distribution among owners, and the rent distribution among renters all line up.
- This matters for the argument: the flow-cost channel runs through payments and rents, so getting those distributions right is what makes the small  $\psi$  credible rather than a fitting artifact.



(a) PTI by LTV



(b) House values, owners



(c) Annual rents, renters

## Backup: where we sit in the literature

### Large penalties, structural:

- Campbell & Cocco (2015) —  $\sim 25\%$  of lifetime consumption.
- Laufer (2018) —  $\sim 29\%$  of permanent income.
- Schelkle (2018) — similar magnitudes.

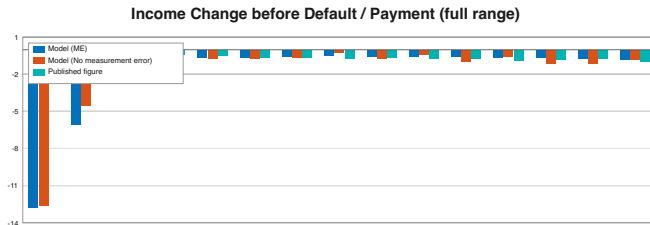
### Liquidity and modest costs, empirical:

- Ganong & Noel — payment vs principal reduction.
- Palmer — IV hazard on house-price exposure.
- Indarte — liquidity vs strategic motives.
- Low; Brueckner et al. — default costs and choice sets.

- Our contribution is not a new estimate of the penalty. It is that **the penalty was never large** once the rent process is right.
- We land on the empirical side of the literature from a fully structural model — which is why the G&N result matters: we can audit their identification inside a world we control.

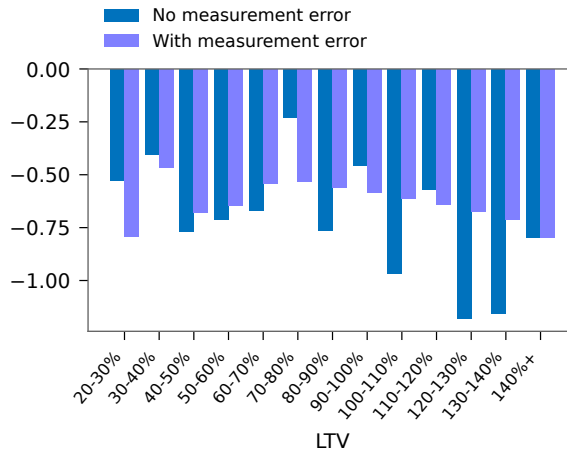
## Backup: $\Delta$ income at default, untruncated

- The published figure truncates at 30% LTV. Below that the ratio explodes:  $-12.8$  and  $-6.1$  in the 0–10% and 10–20% bins.
- Mechanically: these households have collapsed income against a tiny remaining payment, so the denominator vanishes.
- They are under half a percent of defaulters — but they are why the published figure starts at 30%.



## Backup: $\Delta$ income at default, true vs measured LTV

- Same object on raw true LTV rather than operator-weighted bins (paper Fig. 11b).
- On true LTV the profile is **jagged and deeper** — past  $-1$  in the 110–130% bins — because the cells are thin.
- The operator compresses it into a narrow  $-0.4$  to  $-0.8$  band. It **smooths, it does not shift**: the average is the same.
- So the flat income-change profile we show in the body is not an artifact of the smoothing — but it is smoother than the truth.



Average one-year income-change/payment ratio among new defaulters, ages 24–79.

## Backup: fit at the calibrated point

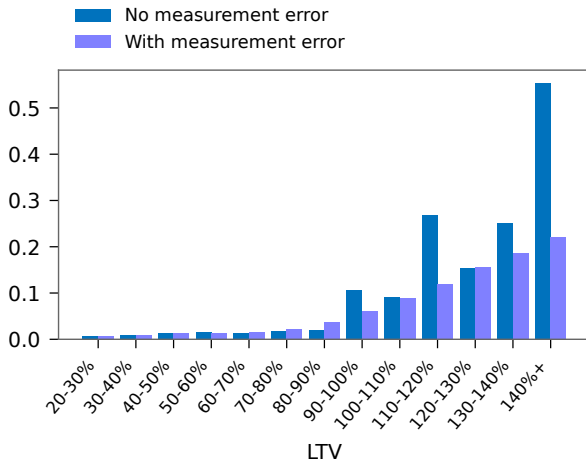
| Moment                       | Model  | Target | Fit term     |
|------------------------------|--------|--------|--------------|
| Homeownership by age         | —      | —      | 0.011        |
| LTV by age                   | —      | —      | 0.041        |
| DTI by age                   | —      | —      | 0.089        |
| PTI by age                   | —      | —      | 0.024        |
| Delinquency by LTV (15 bins) | —      | —      | 0.067        |
| Above-water delinquency /yr  | 0.0211 | 0.0203 | 0.002        |
| Underwater delinquency /yr   | 0.0985 | 0.1012 | 0.001        |
| AW foreclosure   delinq.     | 0.375  | 0.402  | 0.005        |
| <b>Total</b>                 |        |        | <b>0.239</b> |

- Joint eight-parameter search beats the earlier two-parameter winner ( $\psi = 0.14$ ,  $q = 0.0018$ ; total 0.267).
- The gain is in the delinquency block: 0.100  $\rightarrow$  0.074, mostly the bin profile.
- It trades a little of the AW foreclosure aggregate for the AW delinquency level.
- DTI by age (0.089) is the worst-fitting block — the mid-life leverage plateau again.

## On request: how much does measurement error do?

- Observed LTV is noisy

- ▶ Automated valuations: s.d.  $\sigma \approx 14\%$  (Bogin et al.)
- ▶ So some “above water” borrowers are not



Default rate by LTV, same policies and population, no recalibration (paper Fig. 11a).

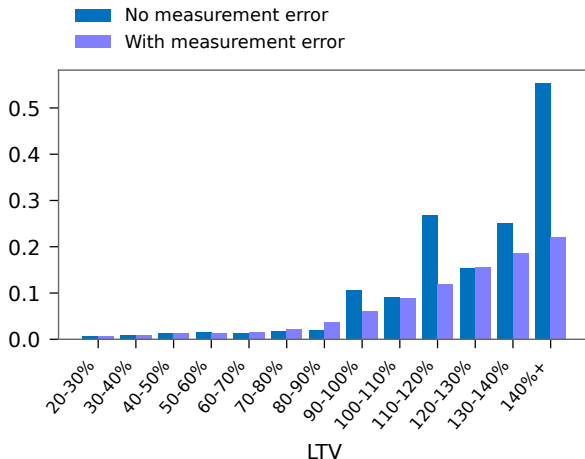
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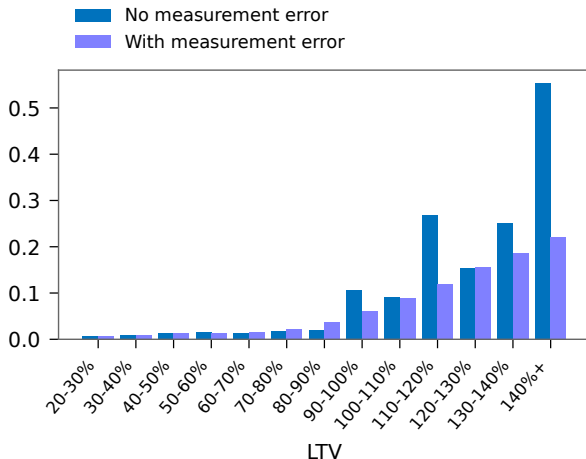
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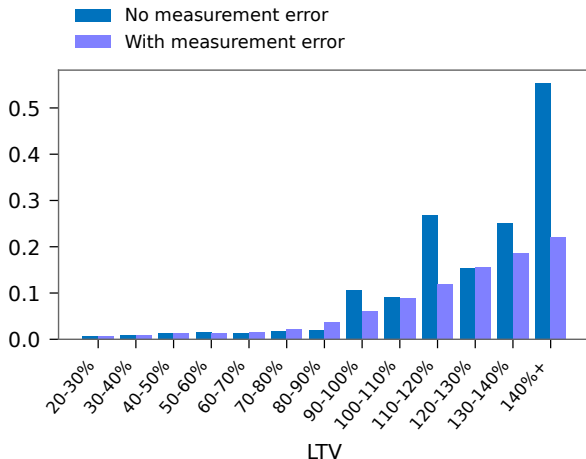
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- **This is load-bearing, and we say so**
  - ▶ True LTV, deep underwater: **0.55** vs 0.27 — over  $2\times$
  - ▶ Measured: **0.22**. That is the 22-vs-27



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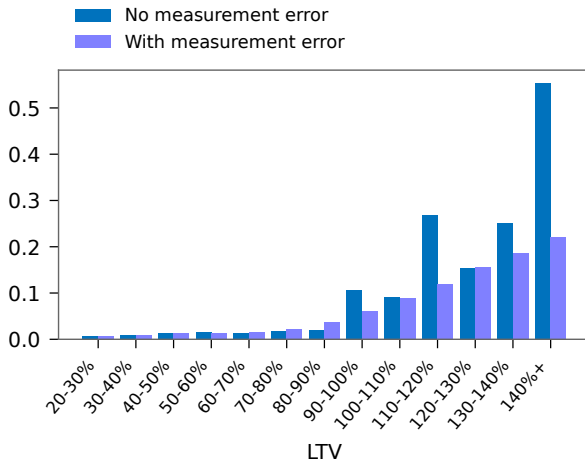
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- **And not the whole story: below 90% the curves coincide**



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